FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2024

TEST DATE: 03-12-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

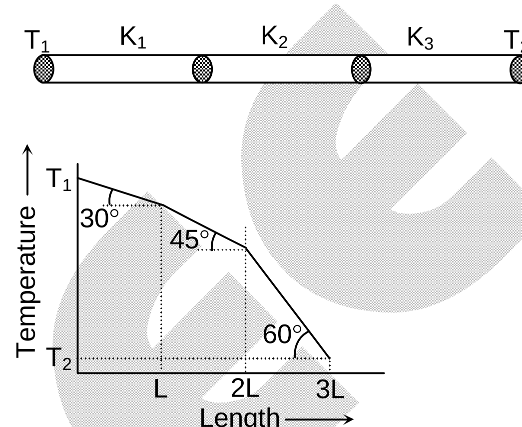
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

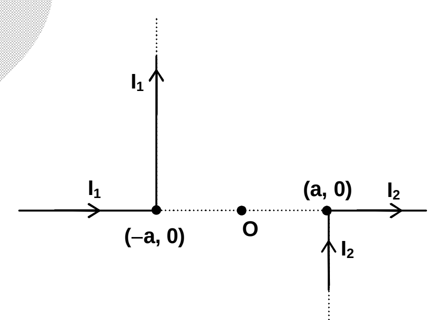
1. Three rods of the same length and the same area of the cross-section are joined. The temperature of two ends are T_1 and T_2 as shown in the figure.
As we move along the rod the variation in temperature is as shown in the following graph. (rods are insulated from the surrounding except at the faces)



(A) $\frac{K_1}{\sqrt{3}} = \frac{K_2}{3} = \frac{K_3}{1}$ (B) $\frac{K_1}{3} = \frac{K_2}{1} = \frac{K_3}{\sqrt{3}}$

(C) $\frac{K_1}{3} = \frac{K_2}{\sqrt{3}} = \frac{K_3}{1}$ (D) $\frac{K_1}{\sqrt{3}} = \frac{K_2}{1} = \frac{K_3}{3}$

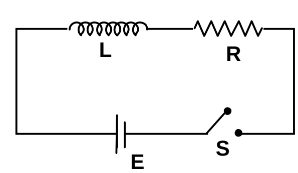
2. Currents I_1 and I_2 flow in the wires as shown in the figure. The magnetic field is zero at distance x to the right of O. Then,



(A) $x = \left(\frac{I_1}{I_2}\right)a$ (B) $x = \left(\frac{I_2}{I_1}\right)a$

(C) $x = \left(\frac{I_1 - I_2}{I_1 + I_2}\right)a$ (D) $x = \left(\frac{I_1 + I_2}{I_1 - I_2}\right)a$

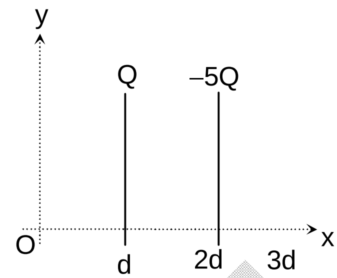
3. In the circuit shown, switch S is closed at time t = 0 sec. The charge that passes through the battery in one time constant is

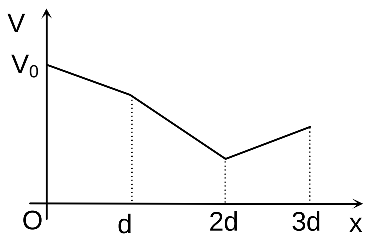


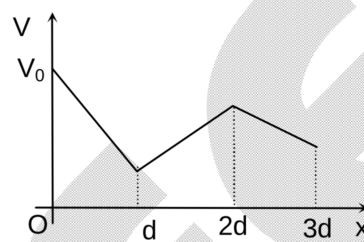
(A) $\frac{eR^2E}{L}$ (B) $E\left(\frac{L}{R}\right)$

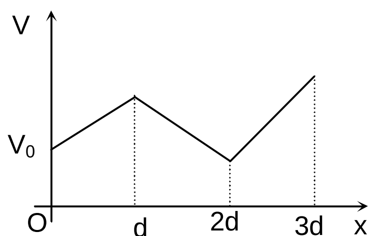
(C) $\frac{EL}{eR^2}$ (D) $\frac{EL}{eR}$

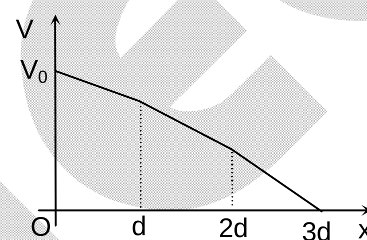
4. Two large identical plates are placed in front of each other at $x = d$ and $x = 2d$ as shown in the figure. If charges on plates are Q and $-5Q$, the potential versus distance graph for the region $x = 0$ and $x = 3d$ is (d is very small and potential at $x = 0$ is V_0)



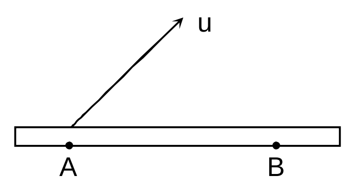
(A)


(B)


(C)


(D)


5. A large flat metal surface has a uniform charge density $(+\sigma)$. An electron of mass m and charge e leaves the surface at the point A with speed u and returns to it at point B. Disregarding gravity the maximum value of AB is

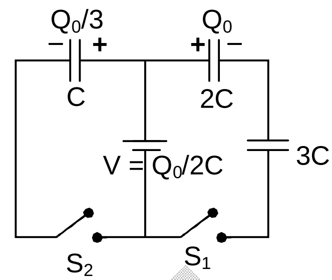


(A) $\dfrac{4u^2m\varepsilon_0}{\sigma e}$
(B) $\dfrac{u^2m\varepsilon_0}{\sigma e}$
(C) $\dfrac{2u^2m\varepsilon_0}{\sigma e}$
(D) $\dfrac{u^2m\varepsilon_0}{2\sigma e}$

6. A non conducting ring of mass m, radius R, having charge q uniformly distributed over its circumference is placed on a rough horizontal surface. A vertical time varying magnetic field $B = 8t^2$ is switched on at time $t = 0$. After 3 sec, the ring starts rotating. The coefiicient of friction between the ring and the table is

(A) $\dfrac{1}{24}\left(\dfrac{qR}{mg}\right)$
(B) $\left(\dfrac{qR}{16mg}\right)$
(C) $\left(\dfrac{16qR}{mg}\right)$
(D) $\left(\dfrac{24qR}{mg}\right)$

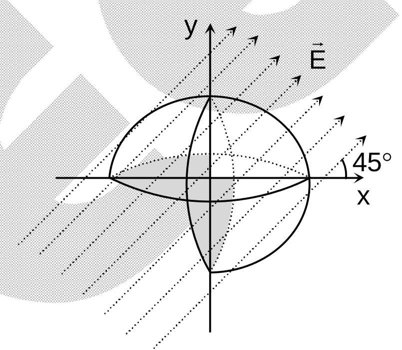
7. In the given circuit, the initial charges on the capacitors are shown in the figure. The charge flown through the switch S_1 and S_2 respectively after closing the switches are



(A) zero, $\frac{Q_0}{6}$ (B) $\frac{Q_0}{5}, \frac{Q_0}{2}$
(C) zero, $\frac{Q_0}{2}$ (D) $\frac{3Q_0}{5}, \frac{Q_0}{6}$

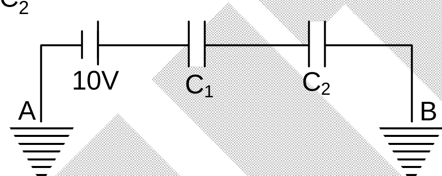
8. A refrigerator absorbs 2000 cal of heat from the ice trays. If the coefficient of performance is 4, the work done by the motor is (1 cal = 4.2 J)
(A) 2100 J (B) 4200 J
(C) 84 00J (D) 500 J

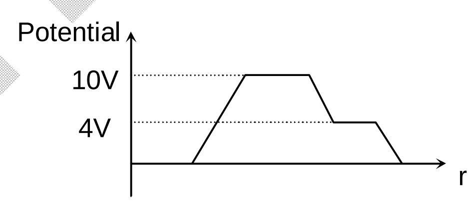
9. One-fourth of a sphere of radius R is removed as shown in the figure. An electric field E exists parallel to the xy plane. Find the flux through the remaining curved part.



(A) $\pi R^2 E$ (B) $\sqrt{2}\pi R^2 E$
(C) $\frac{\pi R^2 E}{\sqrt{2}}$ (D) $\frac{\pi R^2 E}{2\sqrt{2}}$

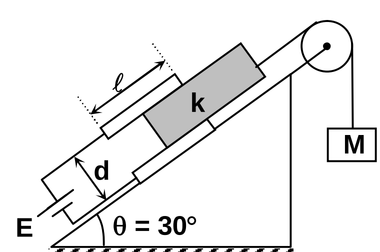
10. Figure shows two capacitors C_1 and C_2 connected with 10V battery and terminal A and B are earthed. The graph shows the variation of potential as one moves from left to right. Then the ratio $\frac{C_1}{C_2}$ is





(A) 5/2 (B) 2/3
(C) 2/5 (D) 4/3

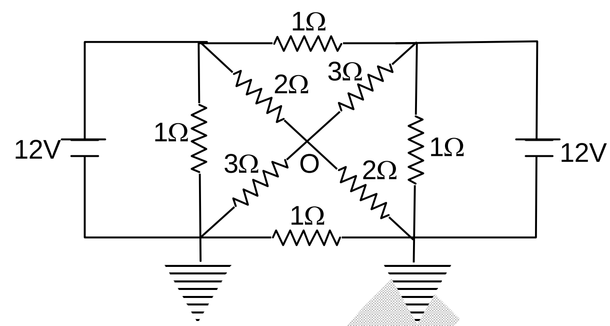
11. The capacitor plates are fixed on an inclined plane and connected to a battery of emf E, The capacitor plates have plate area A, length ℓ and the distance between them is d. A dielectric slab of mass m and dielectric constant k is inserted into the capacitor and tied to mass M by a massless string as shown in the figure. Find value of M for which the slab will stay in equilibrium. There is no friction between slab and plates.



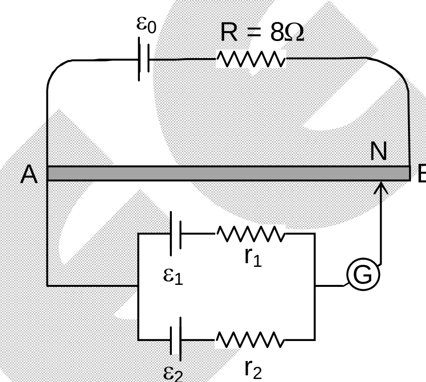
(A) $\frac{m}{2} + \frac{E^2\varepsilon_0 A(k-1)}{2\ell gd}$ (B) $\frac{m}{2} - \frac{E^2\varepsilon_0 A(k-1)}{2\ell gd}$

(C) $\frac{m}{2} + \frac{E^2\varepsilon_0 A(k-1)}{\ell gd}$ (D) $\frac{m}{2} - \frac{E^2\varepsilon_0 A(k-1)}{\ell gd}$

12. The potential of the point O is
(A) 3.5 V
(B) 6.5 V
(C) 3 V
(D) 6V



13. A battery of emf ε_0 = 12V is connected across a 4 m long uniform wire having resistance 4 Ωm^{-1}. The cells of small emf ε_1 = 2V and ε_2 = 4V having internal resistances r_1 = 2Ω and r_2 = 6Ω, respectively, are connected as shown in the figure. If galvanometer shows no deflection at point N, the distance of point N from point A is
(A) 1/6 m
(B) 1/3 m
(C) 25 cm
(D) 50 cm



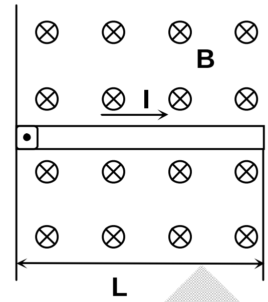
14. The work done in turning a magnet of magnetic moment M by an angle of 90° from the magnetic meridian is n times the corresponding work done to turn it through an angle of 60°, where n is
(A) 1/2 (B) 2
(C) 1/4 (D) 1

15. A paramagnetic material has 10^{28} atom/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is
(A) 3.672×10^{-4} (B) 3.726×10^{-4}
(C) 3.267×10^{-4} (D) 2.672×10^{-4}

16. A step up a transformer operates on a 230 V line and a load current of 2A. The ratio of the primary and secondary winding is 1 : 25. The current in the primary is
(A) 25 A (B) 50 A
(C) 15 A (D) 125 A

17. The value of γ is $\frac{4}{3}$ for an adiabatic process of an ideal gas for which internal energy U = k + nPV, where k is constant. The value of n is
(A) 2 (B) 3
(C) 4 (D) 5

18. A straight conductor of mass m and carrying a current I is hinged at one end and placed in a plane perpendicular to the magnetic field of intensity $\vec{B}$ as shown in the figure. At any moment if the conductor is let free, then the angular acceleration of the conductor will be (neglect gravity)

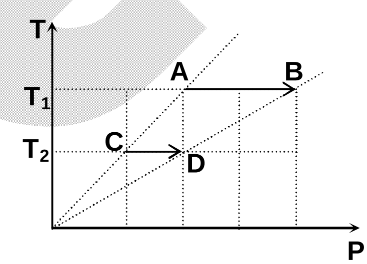


(A) $\frac{IB}{2m}$ (B) $\frac{2IB}{3m}$
(C) $\frac{3IB}{2m}$ (D) $\frac{IB}{4m}$

19. A hollow sphere of radius 2R is charged to a potential V and another smaller sphere of radius R is charged to a potential $\frac{V}{2}$. The smaller sphere is now placed inside the bigger sphere without changing the net charge on each sphere. The potential difference between the two spheres is

(A) $\frac{3V}{2}$ (B) $\frac{V}{4}$
(C) $\frac{V}{2}$ (D) V

20. On a T-P diagram, two moles of an ideal gas perform process AB and CD. If the work done by the gas in the process AB is two times the work done in the process CD, then what is the value of $\frac{T_1}{T_2}$?



(A) 1/2 (B) 1
(C) 2 (D) 4

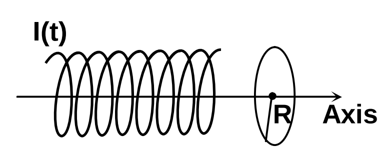
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A charged particle having charge q and mass m accelerated by a potential difference V_0 enters inside a parallel plate capacitor (parallel to length ℓ of plates) at mid point of its separation between plates at t = 0. Potential difference across capacitor varies with time t as V = adt, where a is constant and d is separation between plates. Angle of deviational as it comes out of plates is $\frac{\pi}{4}$. Value of $\frac{ma^2\ell^4}{2qV_0^3}$, will be

22. A solenoid and a ring is placed co-axially as shown in the figure. In solenoid current varies with time as $I = \alpha t^2$. The mutual inductance of the solenoid and ring is M. If the resistance of the ring is R, then current flows through the ring at t = 2 sec is $\frac{X\,M\alpha}{R}$; then X is ……….

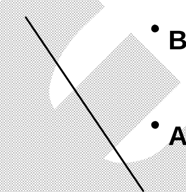


23. A solenoid of length of 0.4 m and having 500 turns of wire carries a current of 3 ampere. A thin circular coil having 10 turns of wire and of radius 0;01 m carries a current of 0.4 ampere. Calculate the torque (in μNm) required to hold the coil in the middle of the solenoid with its axis perpendicular to the axis of the solenoid. (Take $\pi^2 = 10$)

24. An ideal monoatomic gas undergoes a process in which its internal energy U and density ρ vary as $U\rho$ = constant. The ratio of change in internal energy and the work done by gas is (0.5x). Find the value of x.

25. A charged particle having specific charge ratio α CKg^{-1} leaves the origin with a velocity $v_0\hat{i}$, $\vec{E}$ and $\vec{B}$ uniform electric field and magnetic field are directed along y-axis. The value of y-coordinate of the particle when it crosses the y-axis third time is $\frac{x\pi^2 E}{\alpha B^2}$. The value of x is

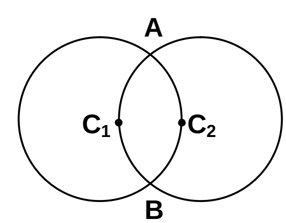
26. An infinite plane of charge with $\sigma = 2\varepsilon_0 \frac{C}{m^2}$ is tilted at a 37° angle to the vertical direction as shown in the figure. Find the potential difference, $V_A - V_B$ in volts, between points A and B at 5 m distance apart. (where B is vertically above A)



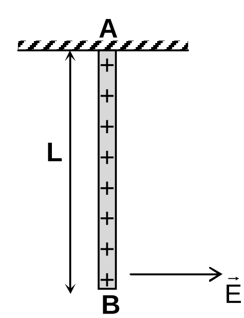
27. A certain mass of a solid exists at its melting temperature of 20°C. When a heat Q is added, $\frac{3}{4}$ of the material melts. When an additional Q amount of heat is added the material transforms to its liquid state at 50°C. Find the ratio of specific latent heat of fusion (in J/g) to the specific heat capacity of the liquid (in $Jg^{-1}°C^{-1}$) for the material.

28. The power radiated by a black body is P, and it radiates maximum energy at the wavelength λ_0. If the temperature of black body is now changed so that it radiates maximum energy at wavelength $\frac{3\lambda_0}{4}$, the power radiated by it is $\frac{nP}{81}$. Then find the value of 'n'.

29. Two circular rings of identical radii and resistance of 36Ω each are placed in such a way that they cross each others centre C_1 and C_2 as shown in figure. Conducting joints are made at intersection points A and B of the rings. An ideal cell of emf 20V is connected across A and B. Find the power delivered by the cell (in watt).



30. A rod AB of length L and mass m is uniformly charged with a charge Q, and is suspended from end A as shown in figure. The rod can freely rotate about A in the plane of the figure. A uniform electric field E is switched on in the horizontal direction due to which the rod gets turned by a maximum angle of 90°. The magnitude of E is equal to $\frac{nMg}{Q}$. Find the value of n.

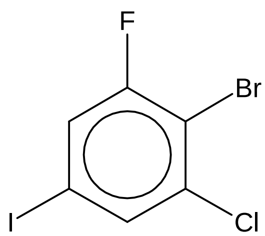


Chemistry PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

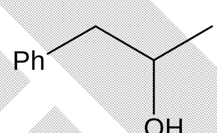
31. The correct IUPAC name of



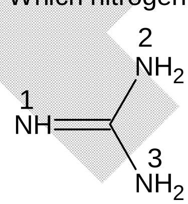
is

(A) 1-Bromo-6-chloro-2-fluoro-4-iodobenzene
(B) 1-Bromo-2-chloro-6-fluoro-4-iodobenzene
(C) 2-Bromo-1-chloro-3-fluoro-5-iodobenzene
(D) 2-Bromo-3-chloro-1-fluoro-5-iodobenzene

32. Which of the following compound gives blood red colouration when its Lassaigne's extract is treated with $FeCl_3$?
(A) Thiourea
(B) Diphenyl sulphide
(C) Phenyl hydrazine
(D) Benzamide

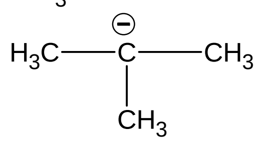
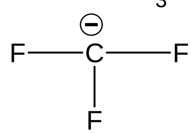
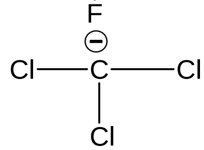
33. An organic compound containing one oxygen gives red colour with ceric ammonium nitrate, decolourises cold dilute alkaline $KMnO_4$, responds iodoform test and shows geometrical isomerism. It should be:
(A) $Ph-CH=CH-CH_2OH$
(B) $Ph-CH=CH-CH(OH)-CH_3$
(C) $Ph-CH=CH-C(=O)-CH_3$
(D) 

34. Which nitrogen is protonated readily in the Guanidine?

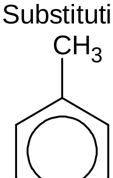


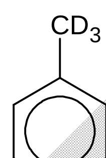
(A) 1
(B) 2
(C) 3
(D) None of these

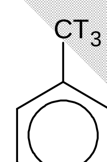
35. If degree of dissociation of benzoic acid, salicyclic acid and 2,6-dihydroxy benzoic acid are given, then it will be found that the degree of dissociation of
(A) Benzoic acid is minimum.
(B) Salicyclic acid is maximum.
(C) 2,6-dihydroxy benzoic acid is minimum.
(D) 2,6-dihydroxy benzoic acid is roughly equal to benzoic acid.

36. Which of the following is most stable?
(A) $\overset{\ominus}{C}H_3$
(B) 
(C) 
(D) 

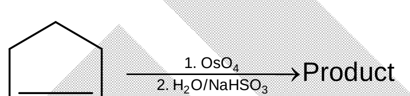
37. Arrange the following in decreasing order of reactivity towards EAS (Electrophilic Aromatic Substitution)


(a)

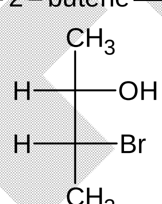
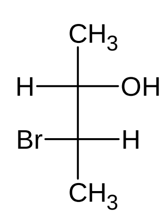
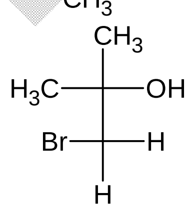
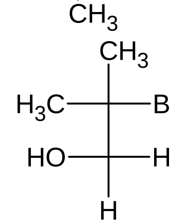

(b)


(c)

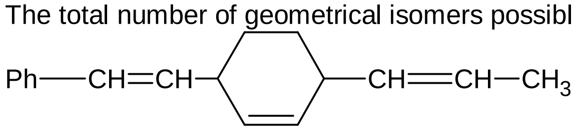
(A) a > b > c
(B) c > b > a
(C) a > c > b
(D) c > a > b

38. 

Product in above reaction would be:
(A) Mixture of Diastereomers
(B) Mixture of Enantiomers
(C) Optically active compounds
(D) Meso compound

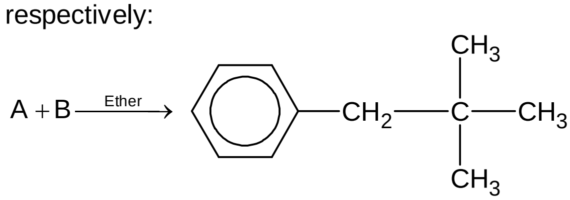
39. $\text{cis}-2-\text{butene} \xrightarrow{\text{HOBr}} P$
(A) 
(B) 
(C) 
(D) 

40. The total number of geometrical isomers possible in following compound is:



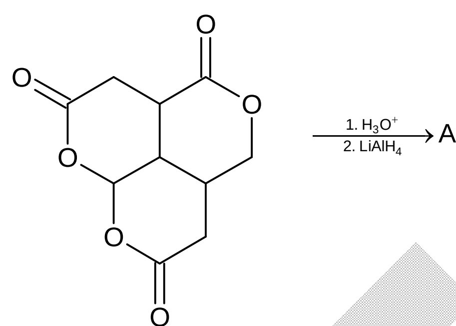
(A) 2
(B) 4
(C) 6
(D) 8

41. The best yield of given product can be obtained by using which set of reactants A and B respectively:

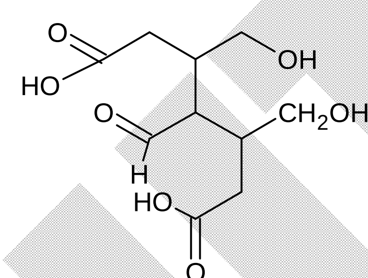


(A) PhLi + Neopentyl chloride
(B) t-BuMgBr + Benzyl bromide
(C) PhMgBr + Neopentyl bromide
(D) Benzyl chloride + t-Butyl chloride

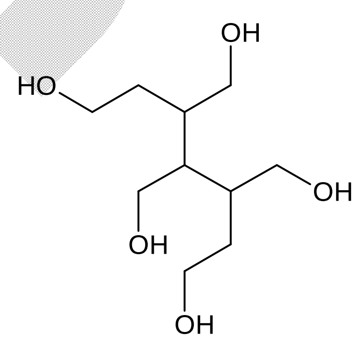
42.



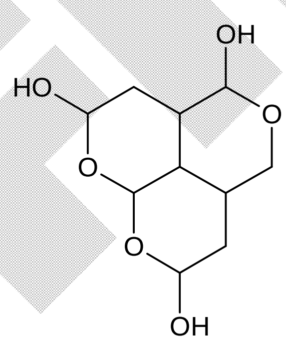
(A)



(B)



(C)



(D) None of these

43. Iodination of benzene is not easily carried out. How can one prepare p-iodobenzoic acid from p-nitro toluene?

(A) (i) $Br_2 + FeBr_3$ (ii) Mg in ether, CO_2 (iii) $3H_2$ / Pt catalyst (iv) HNO_2 at 0^oC (v) KI solution

(B) (i) NBS in CCl_4 and Δ (ii) NaI in acetone (iii) $3H_2$ / Pt catalyst (iv) HNO_2 at 0^oC (v) $H_3PO_2 + H_2O$

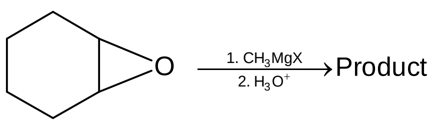
(C) (i) NBS in CCl_4 and Δ (ii) HNO_2 at 0^oC (iii) $CuBr + HBr$ (iv) $KMnO_4 / \Delta$ (v) KI solution

(D) (i) $KMnO_4 / \Delta$ (ii) $Sn + HCl$ (iii) HNO_2 at 0^oC (iv) KI solution

44. Arrange in order of decreasing reactivity towards electrophilic aromatic substitution reaction:

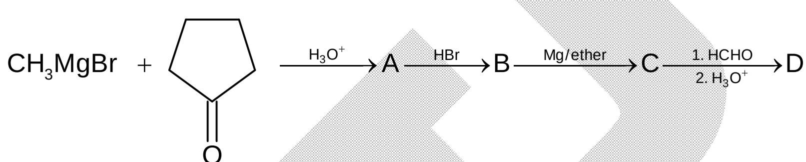
(I) Chlorobenzene
(II) Benzene
(III) Toluene
(IV) Anilinium chloride

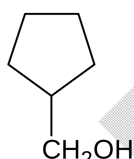
(A) III > II > I > IV
(B) IV > I > II > III
(C) II > I > IV > III
(D) I > II > IV > III

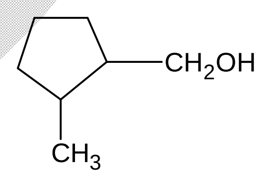
45. 

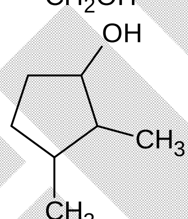
Product is:

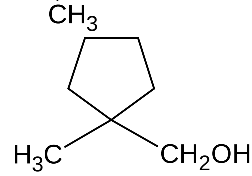
(A) Mixture of enantiomers
(B) Mixture of diastereomers
(C) Meso compound
(D) Achiral

46. 

(A) 

(B) 

(C) 

(D) 

47. Which of the following pair of compounds can be distinguished by aq. $NaHCO_3$?

(A) Phenol and benzyl alcohol
(B) Benzoic acid and picric acid
(C) Picric acid and p-methoxy phenol
(D) Resorcinol and o-cresol

48. The incorrect structure of glycine at given pH are:

(A) $H_3\overset{\oplus}{N}-CH_2-\underset{\underset{O}{\|}}{C}-OH$ at pH = 1.0

(B) $H_3\overset{\oplus}{N}-CH_2-\underset{\underset{O}{\|}}{C}-O^-$ at pH = 6

(C) $H_2N-CH_2-\underset{\underset{O}{\|}}{C}-\overset{\ominus}{O}$ at pH = 10

(D) $H_2N-CH_2-\underset{\underset{O}{\|}}{C}-OH$ at pH = 12

49. Which of the following is not a property of sulphanalic acid?

(A) It is soluble in aq. NaOH
(B) It is soluble in aq. HCl
(C) It is insoluble in organic solvents
(D) All of these

50. Refining of petroleum involves the process of

(A) Simple distillation
(B) Fractional distillation
(C) Steam distillation
(D) Distillation under reduced pressure

SECTION – B

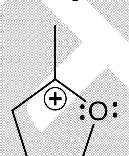
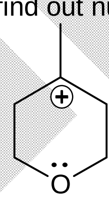
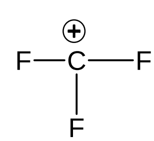
(Numerical Answer Type)

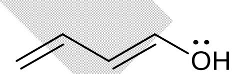
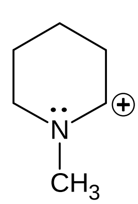
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. How many groups among the following show – I effect?

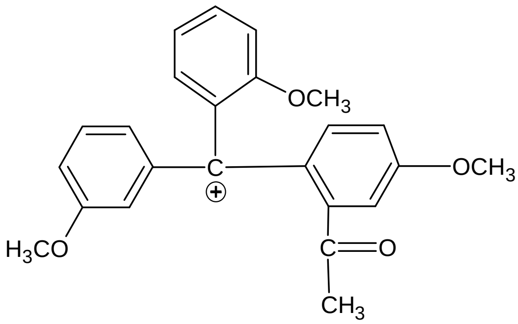
$-CH_3,\ -\overset{\oplus}{N}H_3,\ -OH,\ -\overset{\ominus}{O},\ -N(CH_3)_2,\ -SO_3H,\ -CHO,\ -Cl,\ -COO^-$

52. Among the following, find out number of ions or molecules that can show backbonding?

, , BF_3, 

, , $H_3Si-\ddot{N}=C=S$

53. The positive charge of the following carbocation can be delocalized over how many oxygen atoms in the resonating structure?

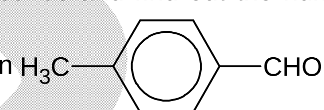


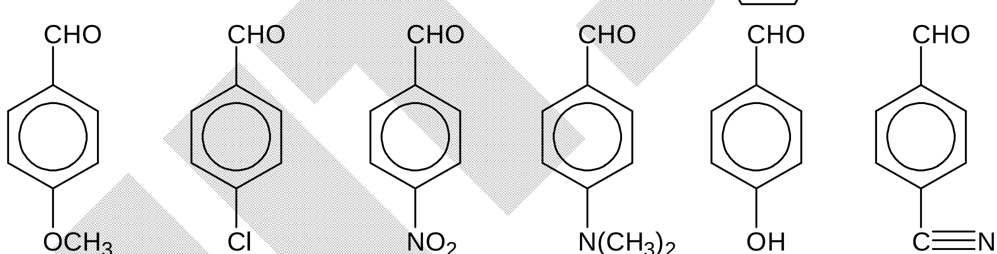
54. Find out the number of dimerize products obtained by following reaction

$$H_3C-Cl + CH_3-CH_2-Cl + CH_3-CH_2-CH_2-Cl \xrightarrow{\text{Dry ether} \mid \text{Na}}$$

55. One mole of 1,2-dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide gave a pentyne. The number of X is:

56. The number of pentyl alcohol (having "pent" word root in IUPAC naming) producing blue colouration in the Victor-Meyer's test is (including stereoisomerism).

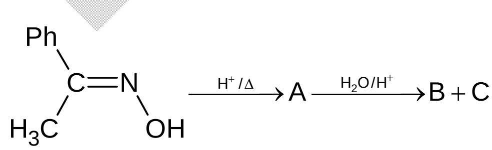
57. Examine the structural formula of following compounds and find out the number of compounds which show higher rate of nucleophilic addition than 



58. $$Ph-NH_2 \xrightarrow[0-5^{o}C]{NaNO_2/HCl} \xrightarrow[\text{dil. HCl}]{Ph-NH_2} \text{Product}(P)$$

If X = Number of nitrogen atoms in the P
Y = Number of stereoisomers of P
Z = Degree of unsaturation of P
Then find sum of X, Y and Z.

59. A decapeptide (mol. Wt. = 796) on complete hydrolysis gives glycine (mol. wt. = 75), alanine and phynylalanine. Glycine contributes 47.0% of the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

60. 

The magnitude of difference in molar mass of B and C is:

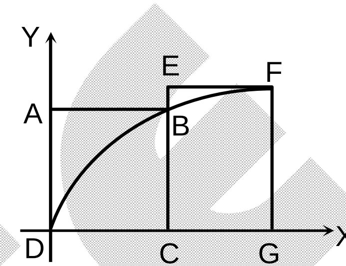
Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is



(A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2}$

(C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{3}+1}{4}$

62. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha, 0)$, $(\beta, 0)$ both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is

(A) $\sqrt{\frac{bc}{a}}$ (B) ac^2

(C) $\frac{b}{a}$ (D) $\sqrt{\frac{c}{a}}$

63. Normal at any point P to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) meets the axes at M and N so that $\frac{PM}{PN}=\frac{2}{3}$, then the value of eccentricity is

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{2}}{\sqrt{3}}$

(C) $\frac{1}{\sqrt{3}}$ (D) none of these

64. The perimeter of a triangle is 14 and two of its vertices are (–3, 0) and (3, 0), then the locus of the third vertex is

(A) $\frac{x^2}{16}+\frac{y^2}{7}=1$ (B) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(C) $\frac{x^2}{7}+\frac{y^2}{16}=1$ (D) $\frac{x^2}{16}+\frac{y^2}{25}=1$

65. If h denotes the arithmetic mean and k denotes the geometric mean of the intercepts made on the axes by a line passing through (1, 1), then the locus of (h, k) is

(A) $y^2 = 2x$ (B) $y^2 = 4x$

(C) $y = 2x$ (D) $x + y = 2xy$

66. The mid points of the sides of a triangle are (5, 0), (5, 12) and (0, 12). The orthocentre of this triangle is
(A) (0, 0)
(B) (10, 0)
(C) (0, 24)
(D) $\left(\frac{13}{3}, 8\right)$

67. Let a, b, c are in G.P. with common ratio r, then the sum of the ordinates of the points of intersection of the line ax + by + c = 0 and the parabola $y^2 = -\frac{x}{2}$ is
(A) $-\frac{r}{2}$
(B) $\frac{r}{2}$
(C) $-\frac{r^2}{2}$
(D) $\frac{r^2}{2}$

68. Parabola $y^2 = 4a(x - c_1)$ and $x^2 = 4a(y - c_2)$ touch each other where c_1 and c_2 are variables, then the locus of their point of contact is
(A) $xy = a^2$
(B) $xy = 2a^2$
(C) $xy = 4a^2$
(D) none of these

69. PQ and QR are two focal chords of an ellipse and the eccentric angles of the points P, Q, R are 2α, 2β and 2γ respectively, then $(\tan\beta).(\tan\gamma) =$
(A) $\cot\alpha$
(B) $\cot^2\alpha$
(C) $2\cot\alpha$
(D) none of these

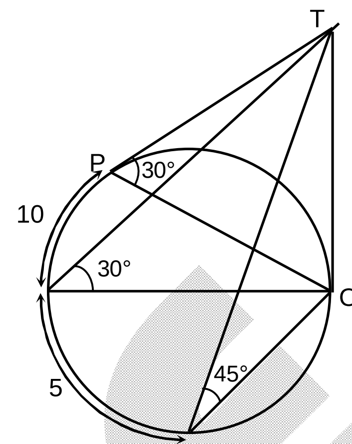
70. If θ be the angle subtended by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at a point (x_1, y_1) and $s_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$, then which is **NOT** correct ?
(A) $\cot\theta = \frac{2\sqrt{s_1}}{\sqrt{g^2 + f^2 - c}}$
(B) $\cot\frac{\theta}{2} = \frac{\sqrt{s_1}}{\sqrt{g^2 + f^2 - c}}$
(C) $\theta = 2\tan^{1} = \sqrt{\frac{g^2 + f^2 - c}{s_1}}$
(D) $\cos\theta = \frac{s_1 + c - g^2 - f^2}{s_1 - c + g^2 + f^2}$

71. For any real k the circle $x^2 + y^2 + 2kx + 2ky - 8 = 0$ passes through two fixed points A and B. Locus of the point of intersection of the tangents to the circle at A and B is
(A) $x^2 = 4y$
(B) $y^2 = 4x$
(C) $x = y$
(D) $x + y = 0$

72. Which of the following is not satisfied by the point of intersection of lines $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$
(A) $x - y = 0$
(B) $(x + y)(a + b) = 2ab$
(C) $(lx + my)(a + b) = (l + m)ab$
(D) $(lx - my)(a + b) = 2(l - m)ab$

73. A straight line is drawn parallel to the conjugate axis of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which meets the hyperbola and its conjugate hyperbola at P and Q respectively. The meeting points of the normals at P and Q to the respective curves always lies on
(A) $x = 0$
(B) $y = 0$
(C) $x + y = 0$
(D) $x - y = 0$

74. A diving stand (vertical) is situated at an end of a circular swimming pool. At a point P on the circular edge the angle of elevation of top of the diving stand is 30°. After moving 10 meters along the circular edge the angle of elevation is again 30° and further going along edge 5 meter. The angle of elevation is 45°, then the perimeter of the pool is
(A) 30 meter
(B) 45 meter
(C) 60 meter
(D) 15 meter



75. For $\alpha \in (0, \pi)$ and $\theta \in (0, \pi)$ the number of values of θ satisfying the equation $\sin 3\alpha = 4\sin\alpha \sin(x + \alpha) \sin(x - \alpha)$ is
(A) 1 (B) 2
(C) 3 (D) 4

76. If $\frac{\cos\theta}{a} = \frac{\sin\theta}{b}$, then $\frac{a}{\sec 2\theta} + \frac{b}{\text{cosec} 2\theta}$ is equal to
(A) a (B) b
(C) a + b (D) $a^2 + b^2$

77. If $\cos(\theta - \alpha) = a$ and $\sin(\theta - \beta) = b$, $\left(0 < \theta - \alpha, \theta - \beta < \frac{\pi}{2}\right)$ then $\cos^2(\alpha - \beta) + 2ab \sin(\alpha - \beta)$ is equal to
(A) $4a^2b^2$ (B) $a^2 - b^2$
(C) $a^2 + b^2$ (D) $-a^2b^2$

78. If $x \in \left[\frac{1}{2}, 1\right]$, then $\cos^{-1} x + \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\left(\sqrt{3 - 3x^2}\right)\right) =$
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

79. Let $x = \cot^{-1}\sqrt{\cos\alpha} - \tan^{-1}\sqrt{\cos\alpha}$, then sinx =
(A) $\tan^2\alpha$ (B) $\cot^2\alpha$
(C) $\tan^2\frac{\alpha}{2}$ (D) $\cot^2\frac{\alpha}{2}$

80. If in ΔABC a, b, c are the length of the sides opposite to angles A, B and C then $\frac{\cot(A/2) + \cot(B/2) + \cot(C/2)}{\cot A + \cot B + \cot C} =$
(A) $\frac{(a + b + c)^2}{a^2 + b^2 + c^2}$ (B) $\frac{a + b + c}{a^2 + b^2 + c^2}$
(C) $\frac{a + b + c}{a^3 + b^3 + c^3}$ (D) $\frac{a^2 + b^2 + c^2}{(a + b + c)^2}$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. The number of points whose both coordinates are integers and lying inside all the three circles $x^2 + y^2 = 16$, $x^2 + y^2 - 8x = 0$ and $x^2 + y^2 - 8y = 0$ are

82. If two parallel lines L_1 and L_2 with positive slopes are tangent to the circle $C_1 : x^2 + y^2 - 2x - 16y + 64 = 0$. If L_1 is also tangent to circle $C_2 : x^2 + y^2 - 2x + 2y - 2 = 0$ and the equation of L_2 is $a\sqrt{a}x - by + c - a\sqrt{a} = 0$ where a, b, c $\in$ N, then the value of $\frac{a+b+c}{2}$ =

83. A moving parabola of latus rectum l touches a fixed equal parabola. The axes of both the curves being parallel. The locus of the vertex of the moving parabola is a parabola of latus rectum kl, then k is equal to

84. An isosceles triangle ABC is inscribed in the circle $x^2 + y^2 = 16$. A = (0, 4) from points A, B and C lines perpendicular to x-axis are drawn which cuts the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at the points P, Q, R respectively such that A and P are on the opposite sides of the x-axis while Q and B are on the same side of x-axis and C and R are also on same side of the x-axis. If ΔPQR is a right angled triangle with $\angle P = 90°$ and θ is the smallest angle of ΔABC then $16\tan^2\theta + 8\tan\theta + 9$ is equal to

85. If a tangent of slope 2 of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then the maximum value of ab is

86. If m is the slope of common tangent of the curves $y = x^2 - x + 1$ and $y = x^2 - 3x + 1$, then |m| =

87. If α, β, γ are the length of medians of ΔABC, then $4\left(\frac{\alpha^2+\beta^2+\gamma^2}{a^2+b^2+c^2}\right)$ =

88. The general solution of the equation $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is $\frac{n\pi}{2}+\frac{\pi}{p}$, n $\in$ I, then P =

89. If $\sin 5\theta = a\sin^5\theta + b\sin^3\theta + c\sin\theta + d \ \forall \ \theta \in R$, then a + b + c + d =

90. The number of solutions of equation $\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}=0$ in the interval $\left(0,\frac{\pi}{4}\right)$ is